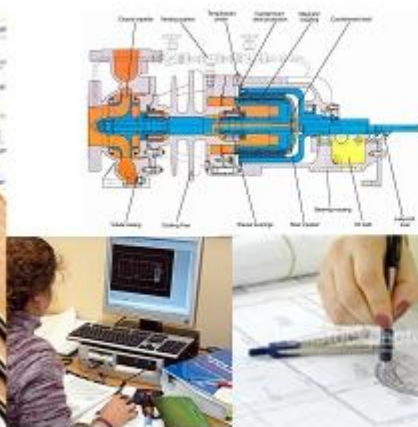


Orthographic Projection



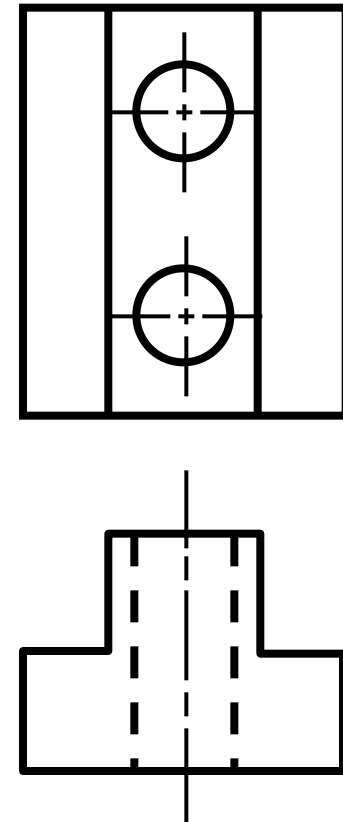
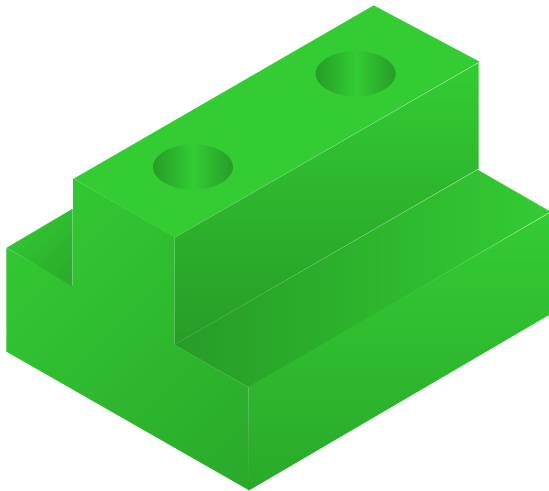
TOPICS

- Object representation
- Multiview projection
- Glass box concept
- Orthographic projection of point, line, plane, surface and object.
- Line convention

OBJECT REPRESENTATION

■ Axonometric projection

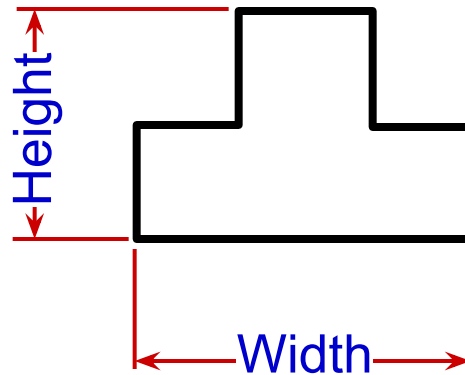
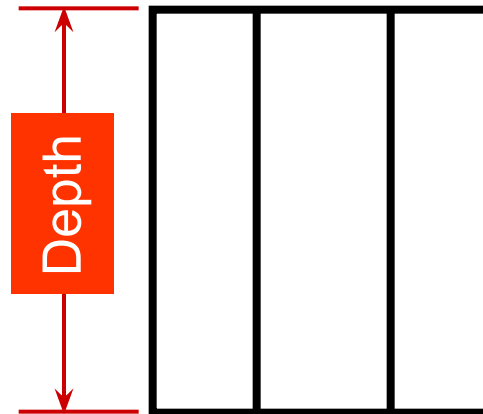
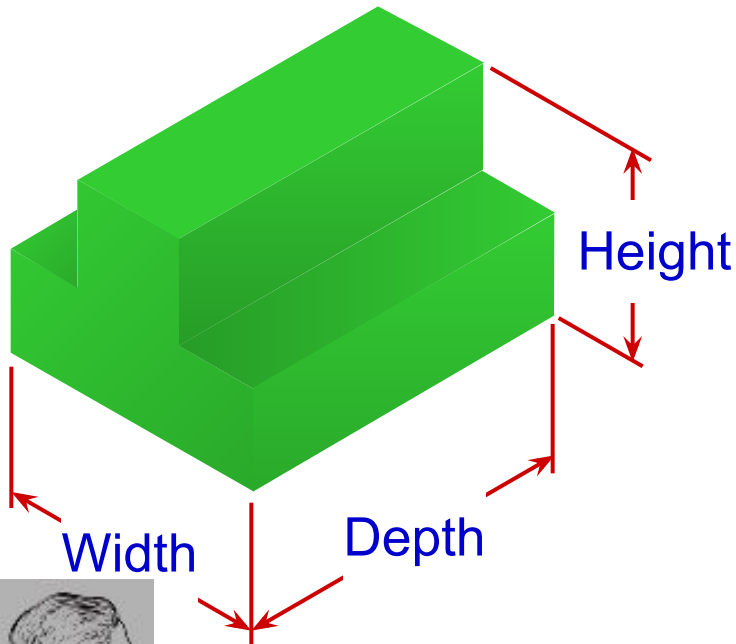
■ Multiview projection



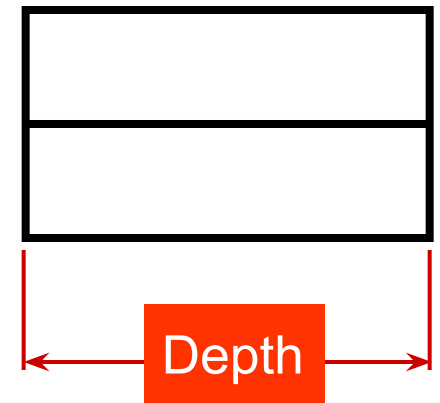
MULTIVIEW PROJECTION

Three principle dimensions
of an object ...

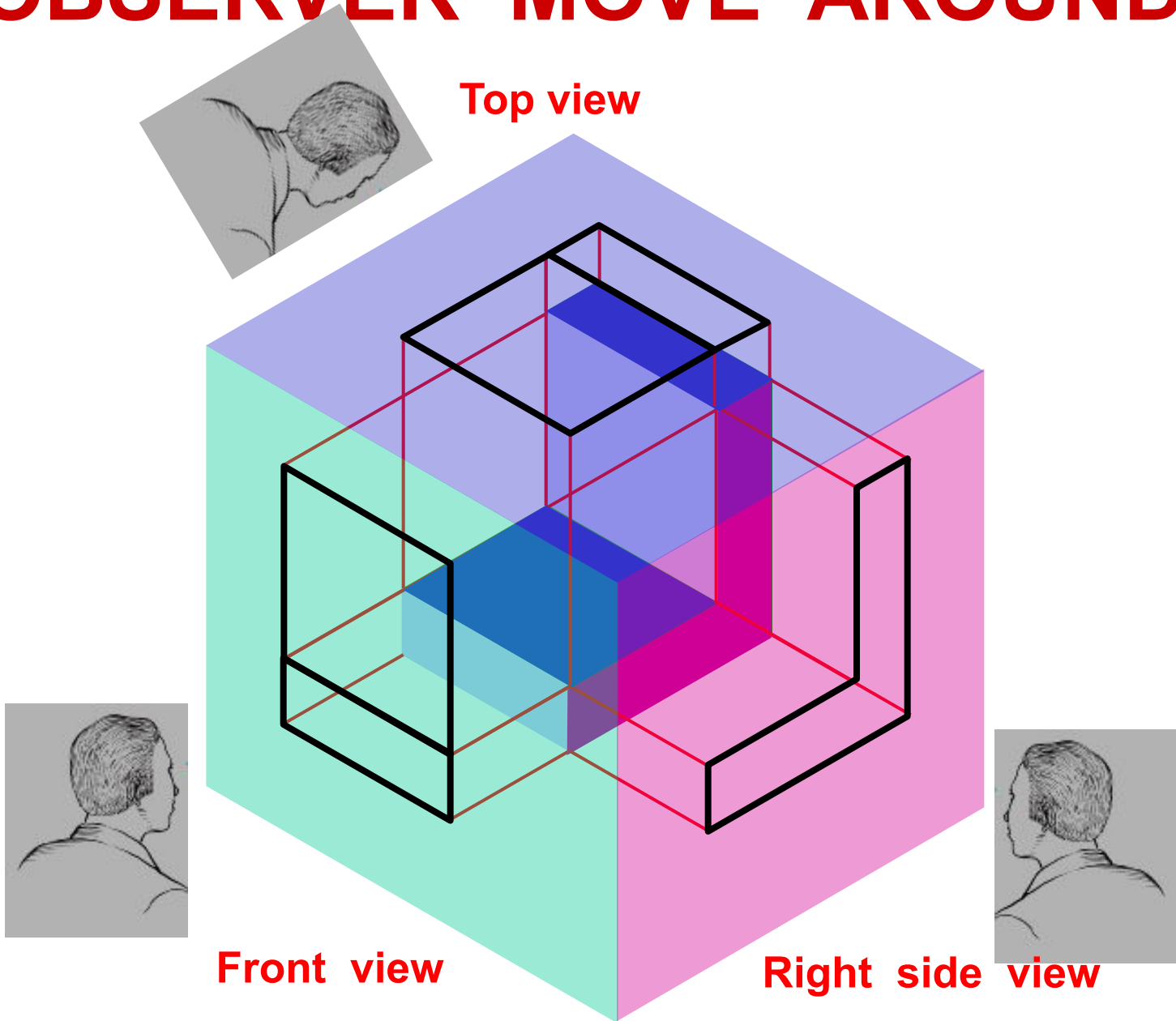
... can be presented only
two in each view.



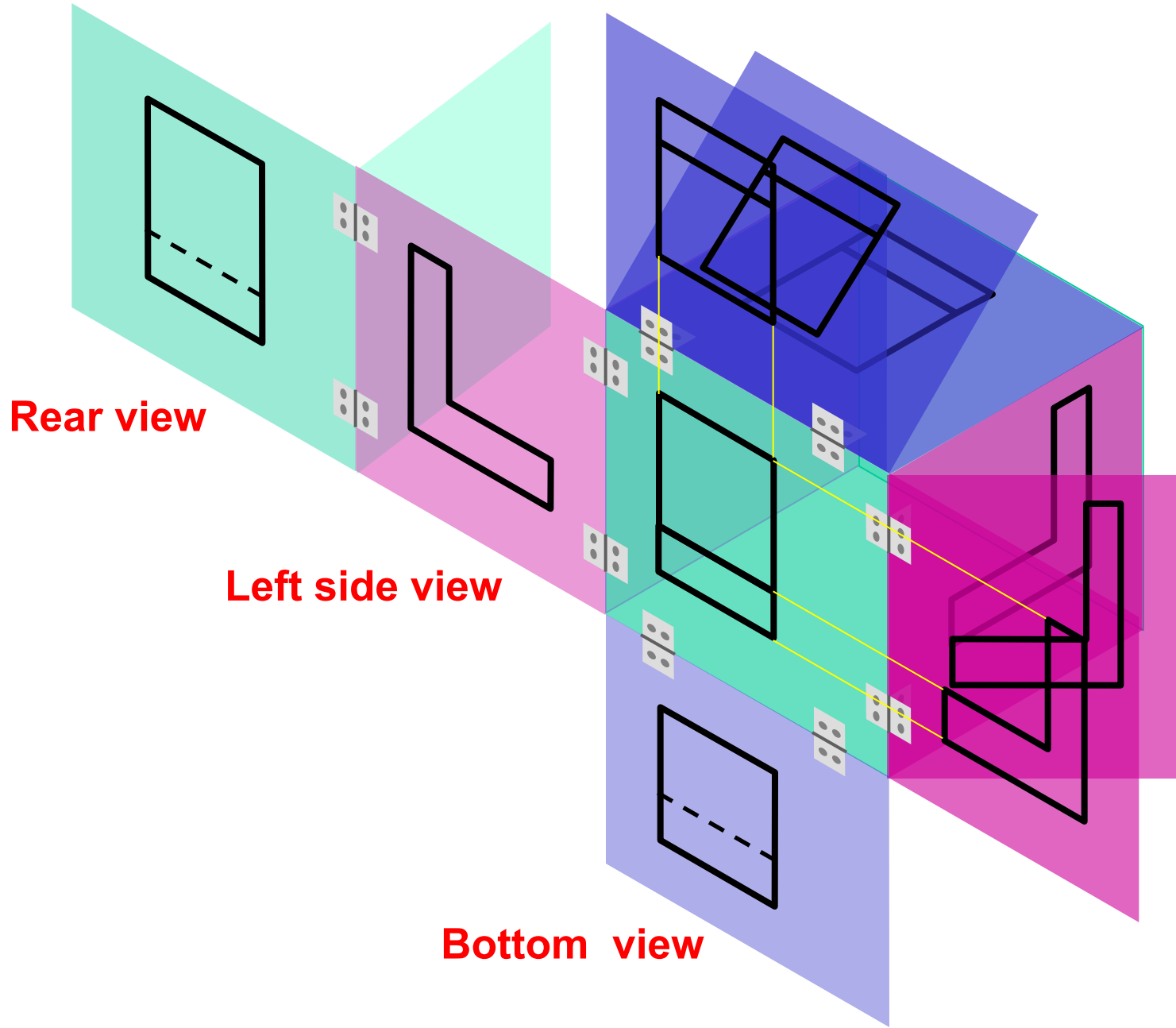
**Adjacent view(s)
is needed to
fulfill the size
description.**

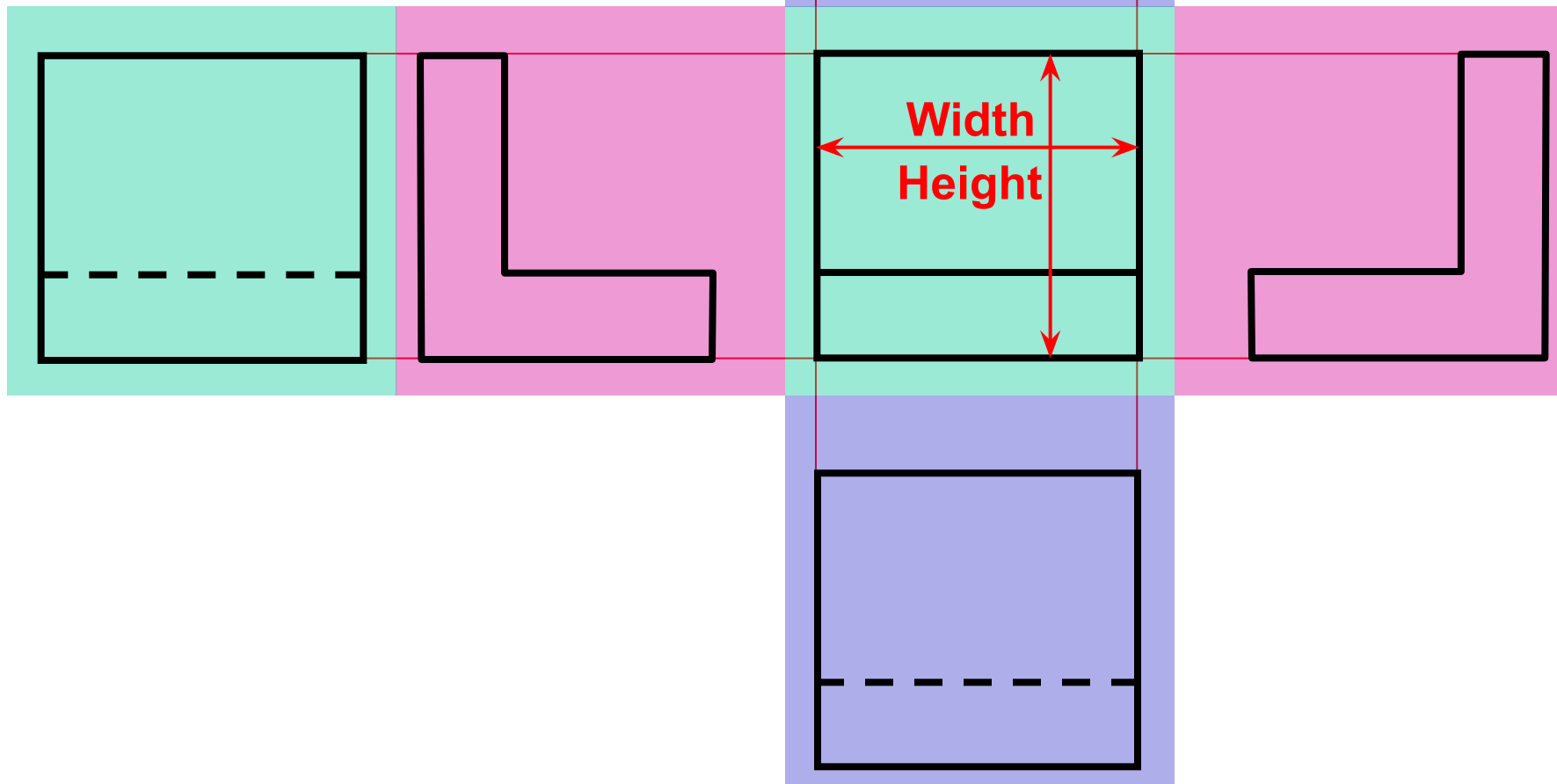
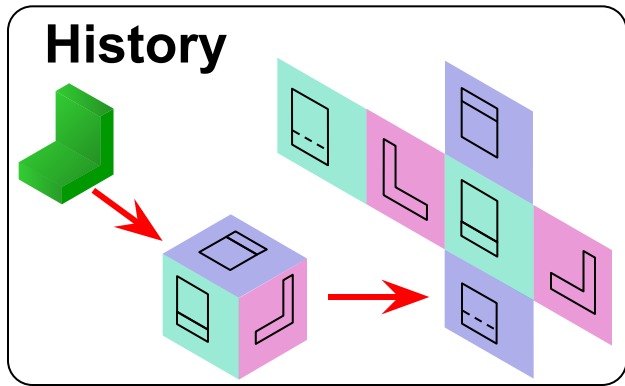


OBSERVER MOVE AROUND



THE GLASS BOX CONCEPT





MULTIVIEW PROJECTION



Orthographic Projection of Object Features



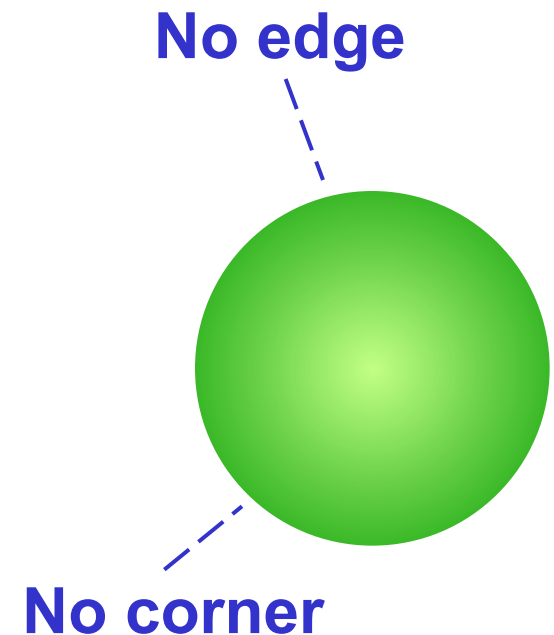
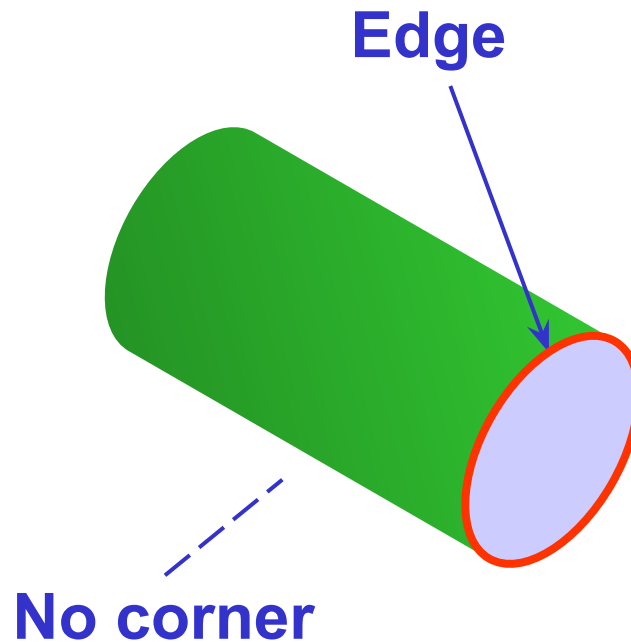
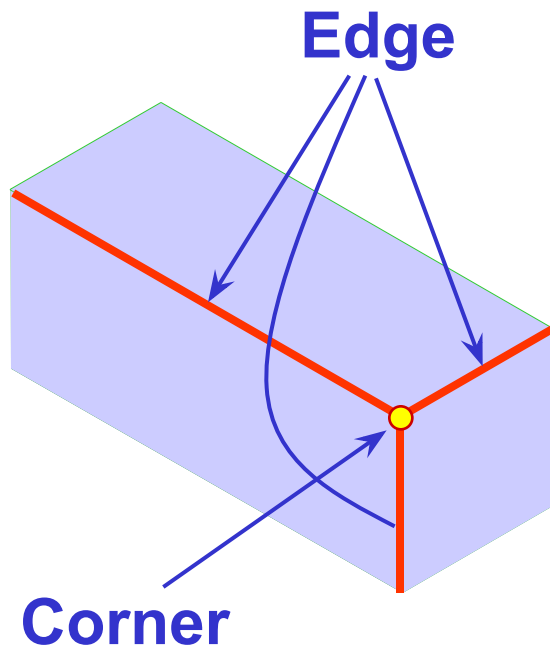
OBJECT FEATURES

Edges

are lines that represent the boundary between two faces.

Corners

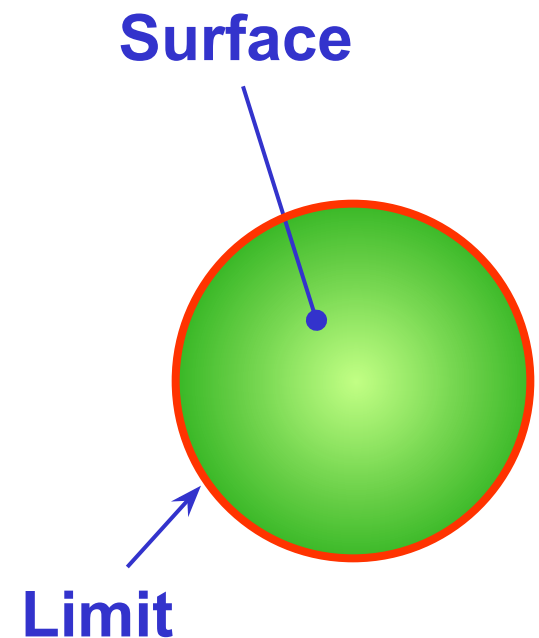
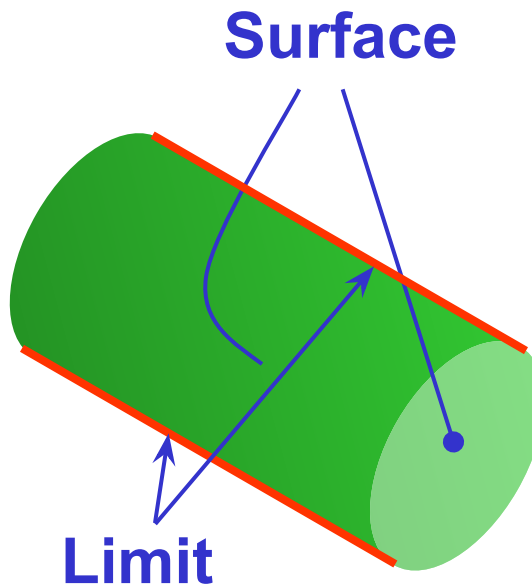
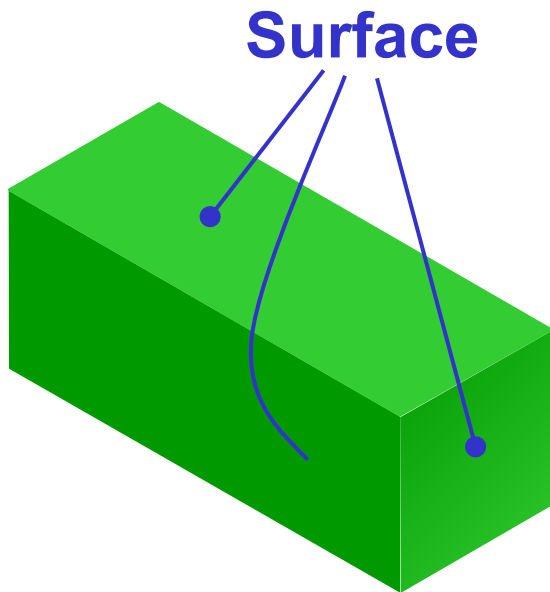
Represent the intersection of two or more edges.



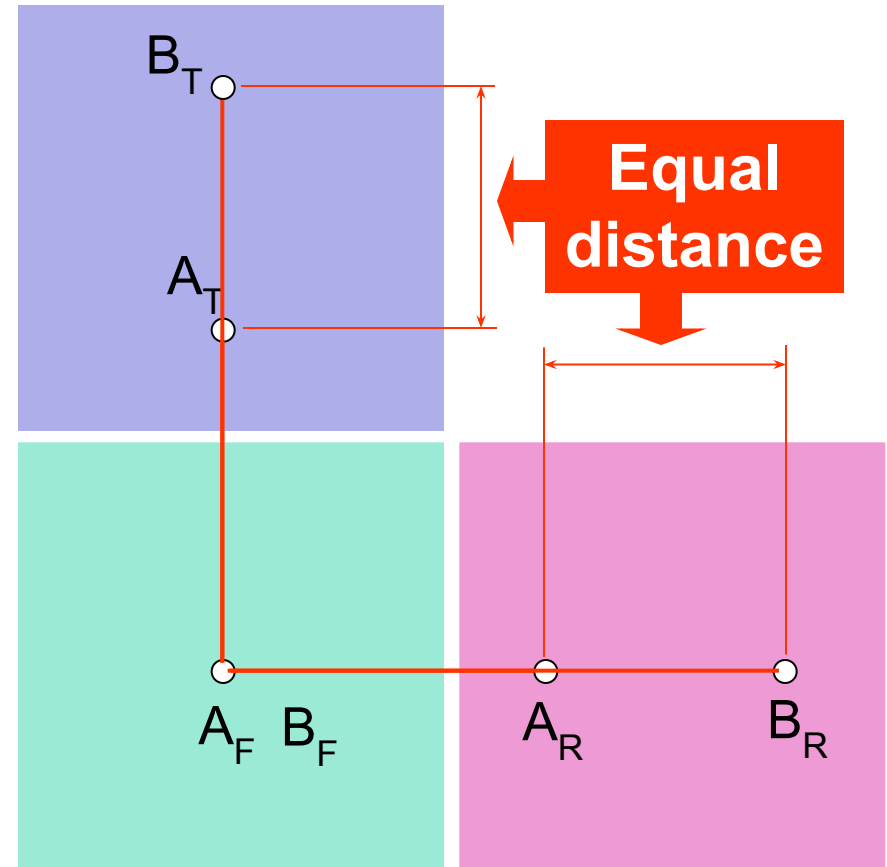
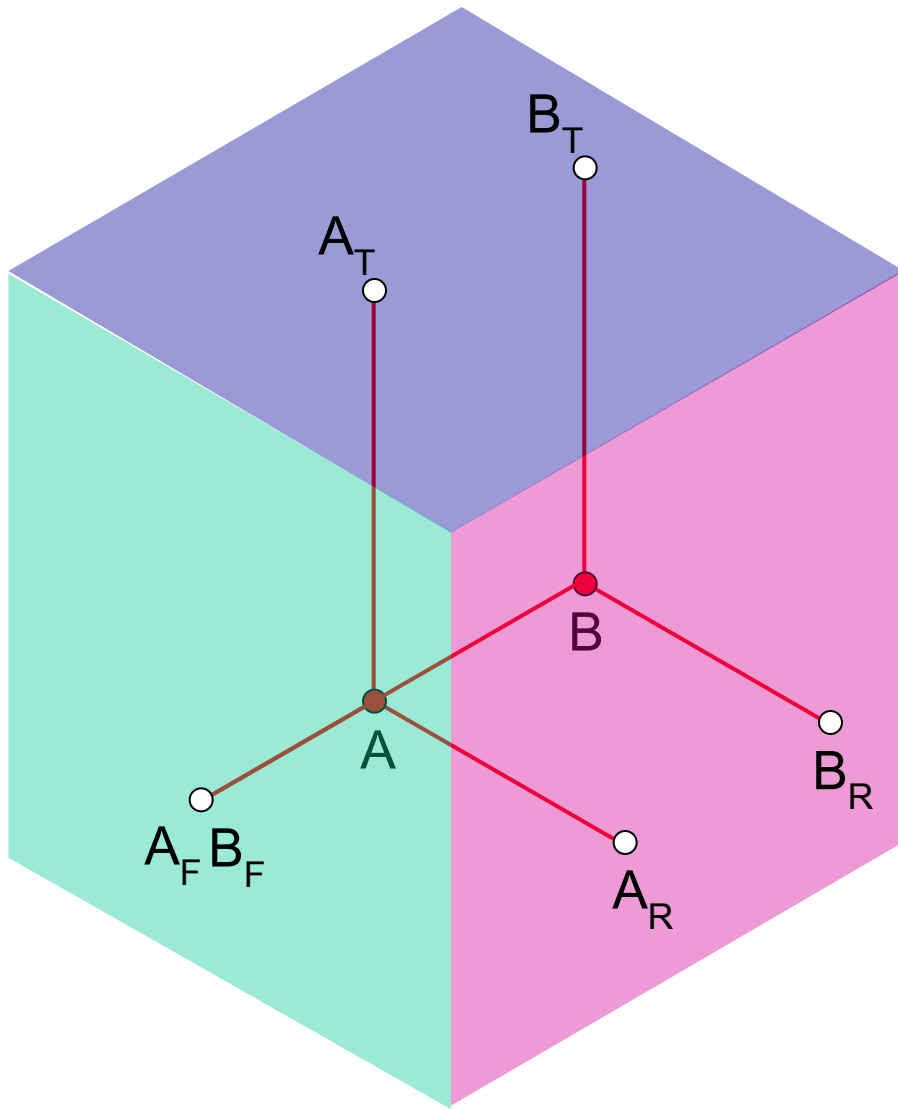
OBJECT FEATURES

Surfaces are areas that are bounded by edges or limiting element.

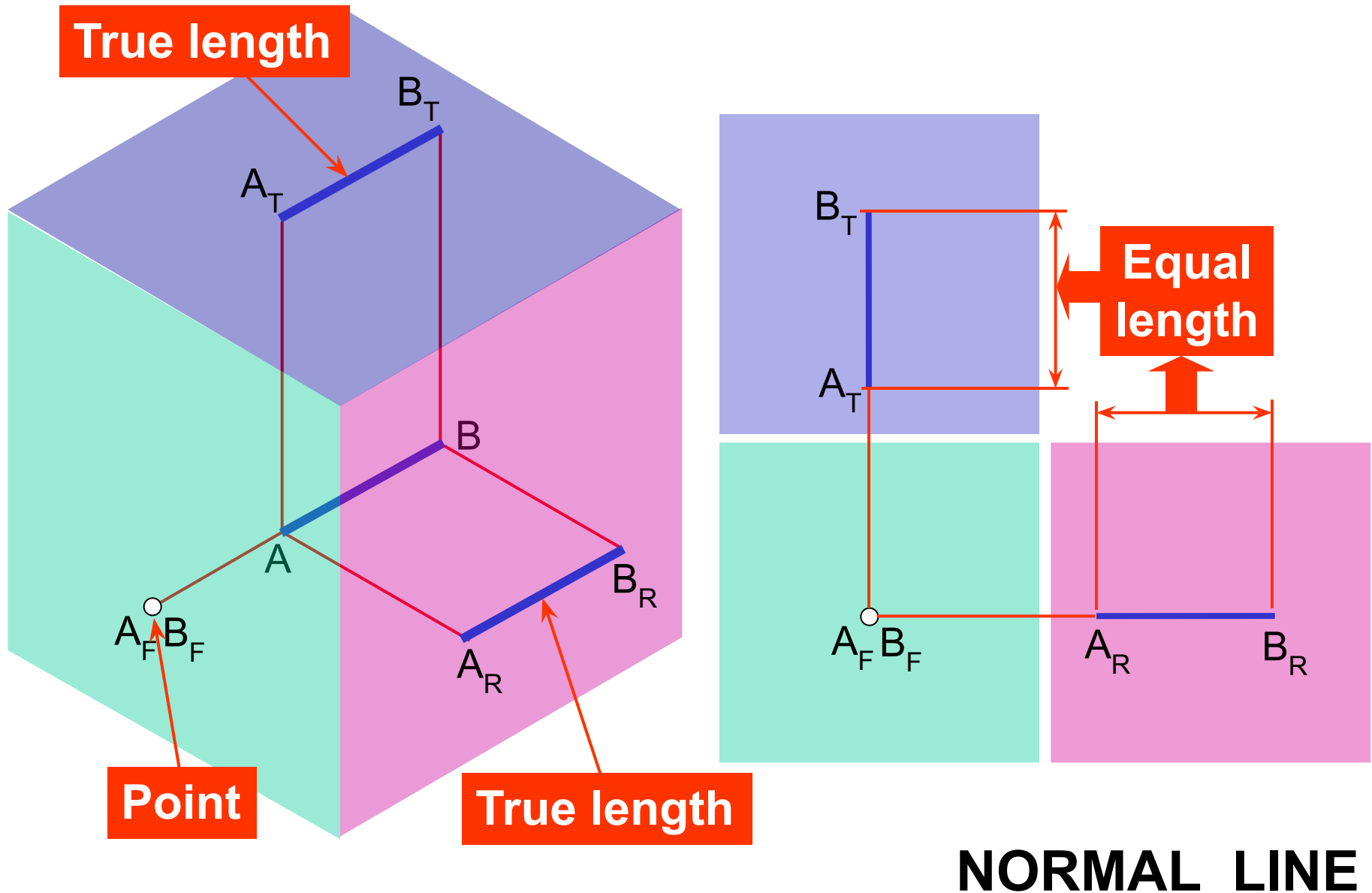
Limiting element is a line that represents the last visible part of the curve surface.



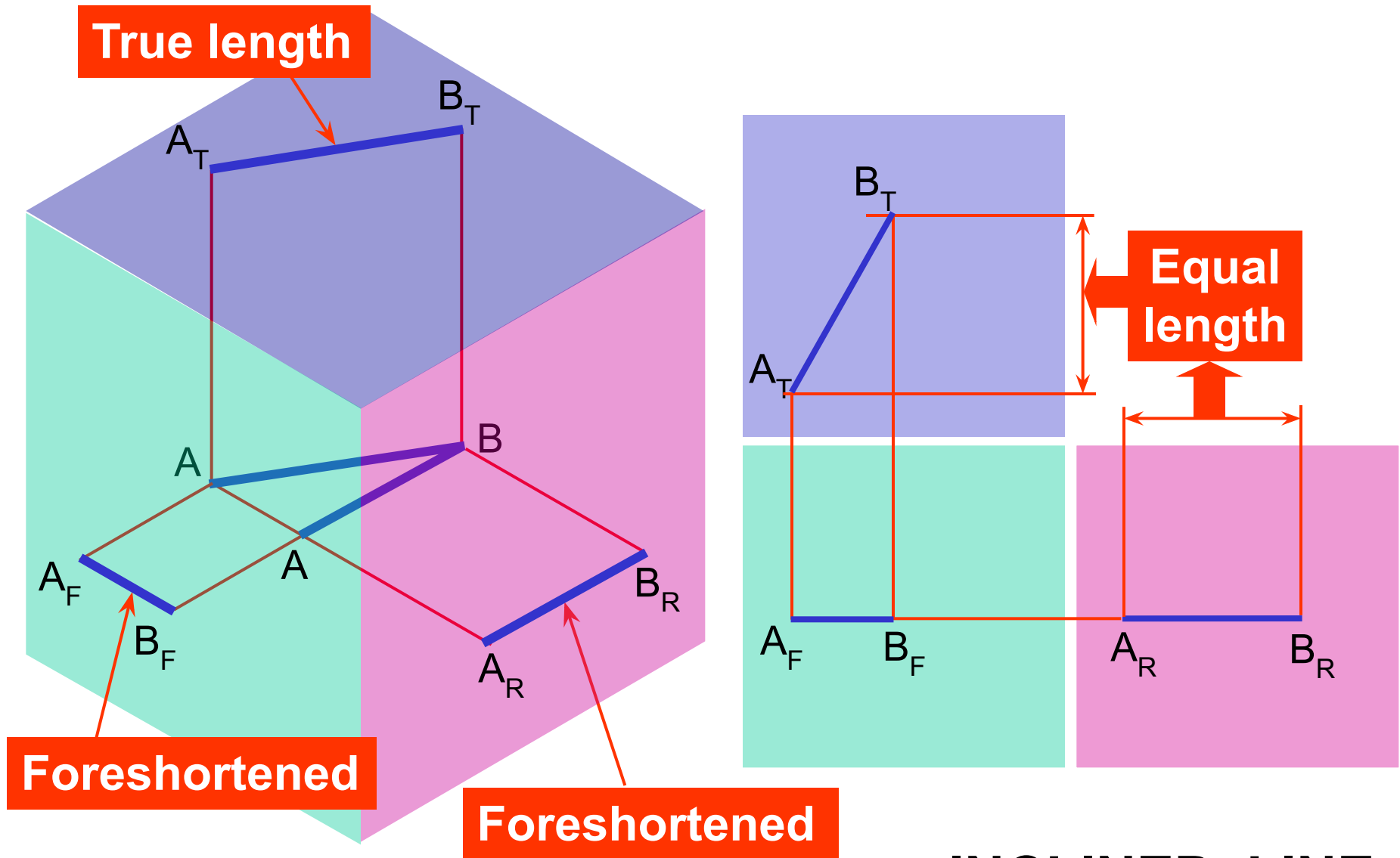
PROJECTION OF POINT(S)



PROJECTION OF LINE



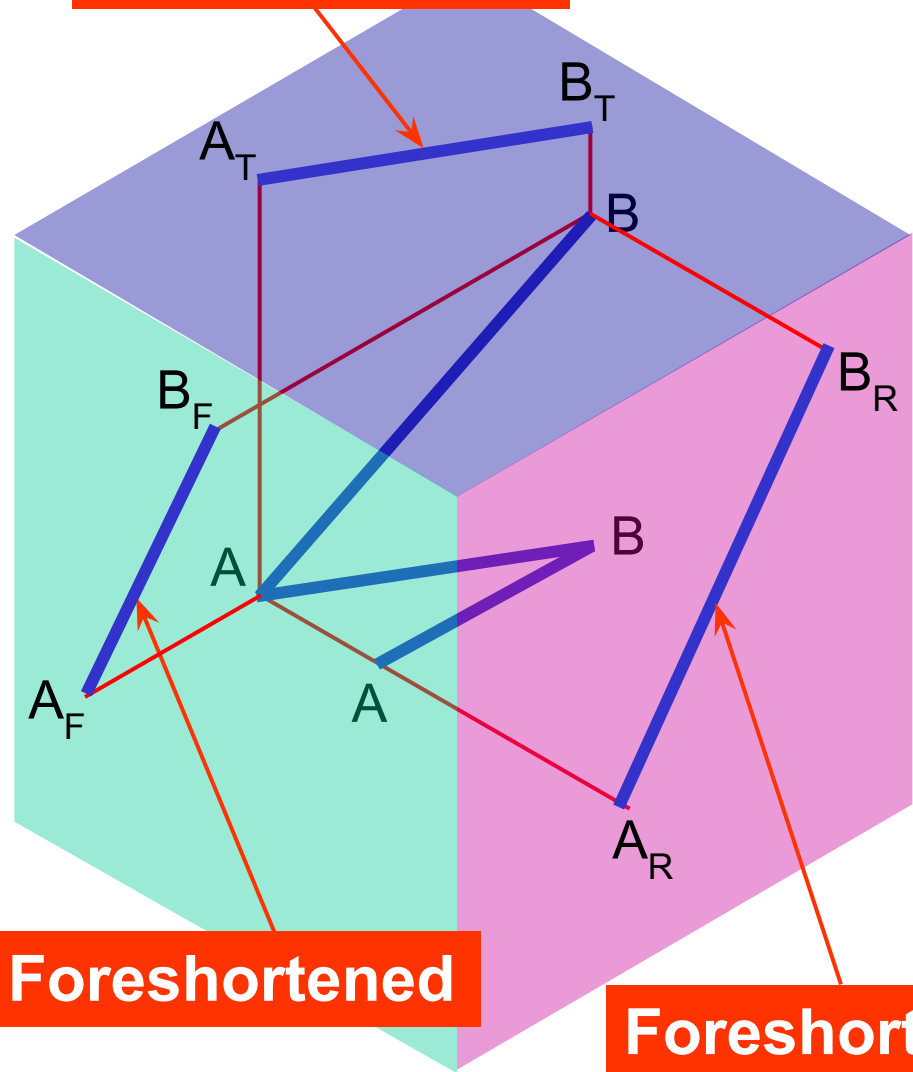
PROJECTION OF LINE



INCLINED LINE

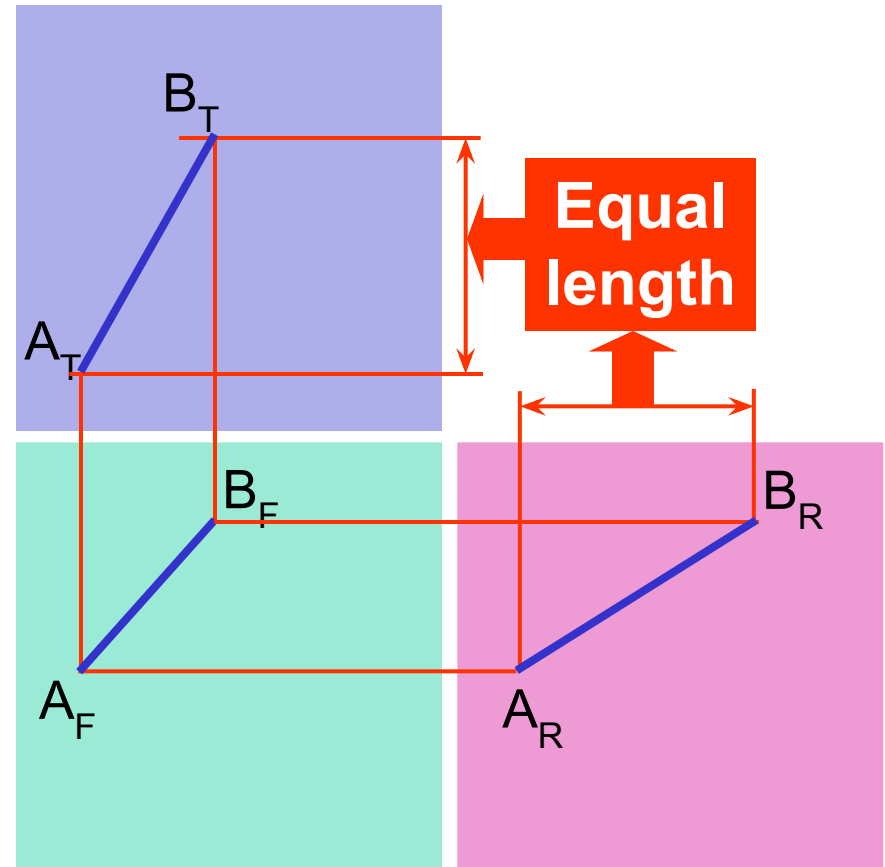
PROJECTION OF LINE

Foreshortened



Foreshortened

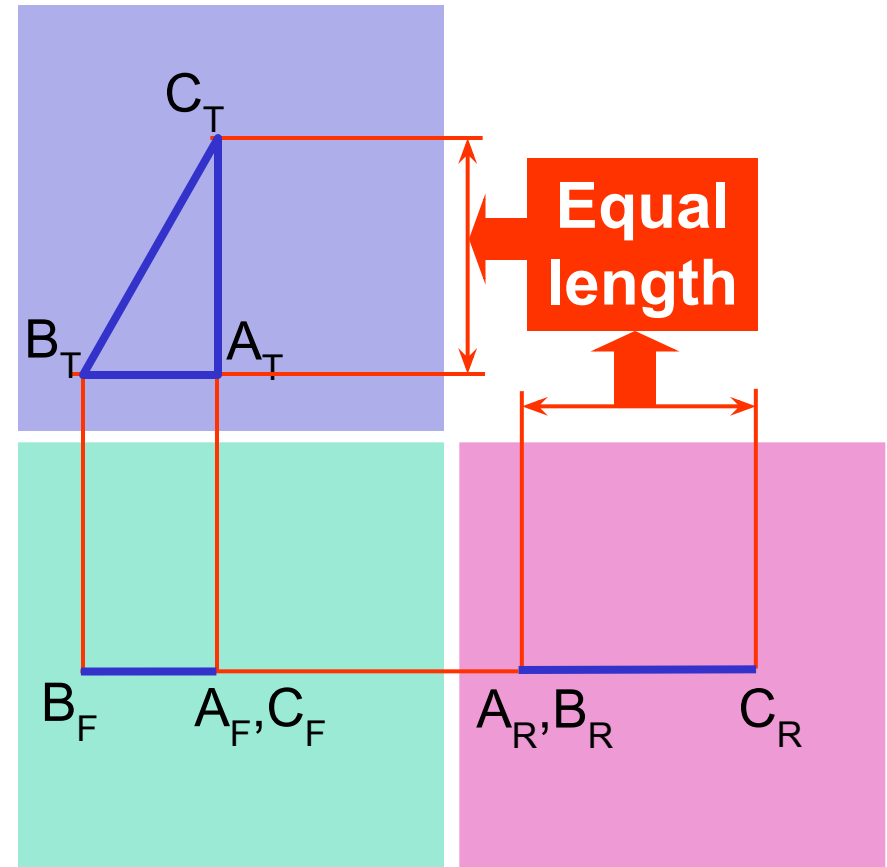
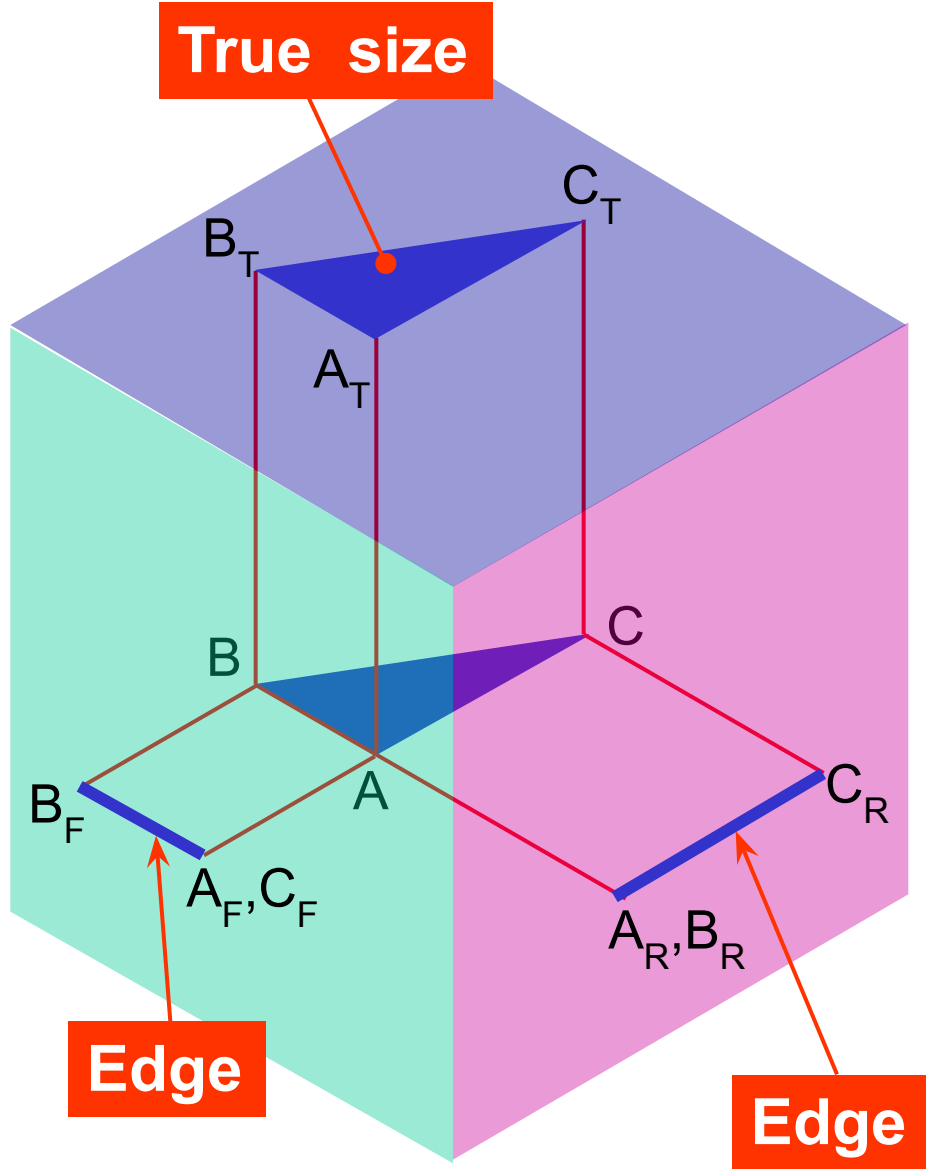
Foreshortened



**Equal
length**

OBLIQUED LINE

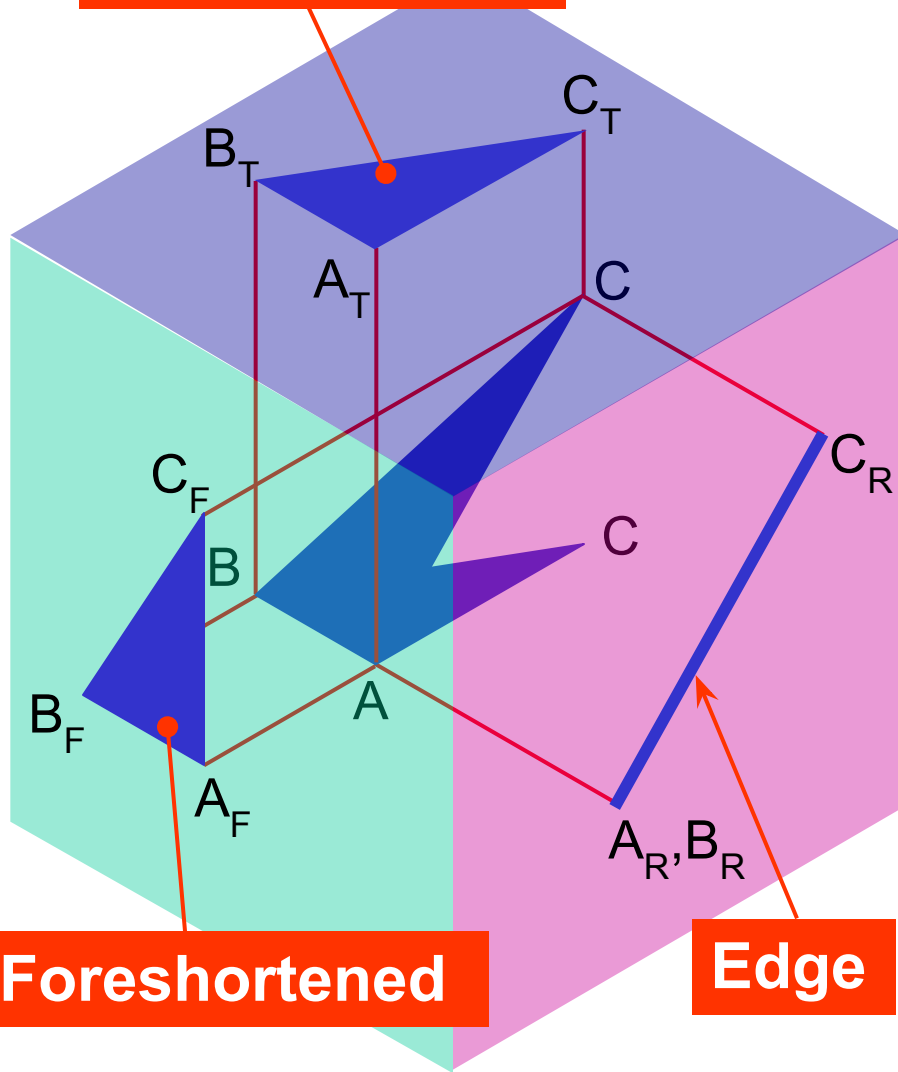
PROJECTION OF PLANE



NORMAL PLANE

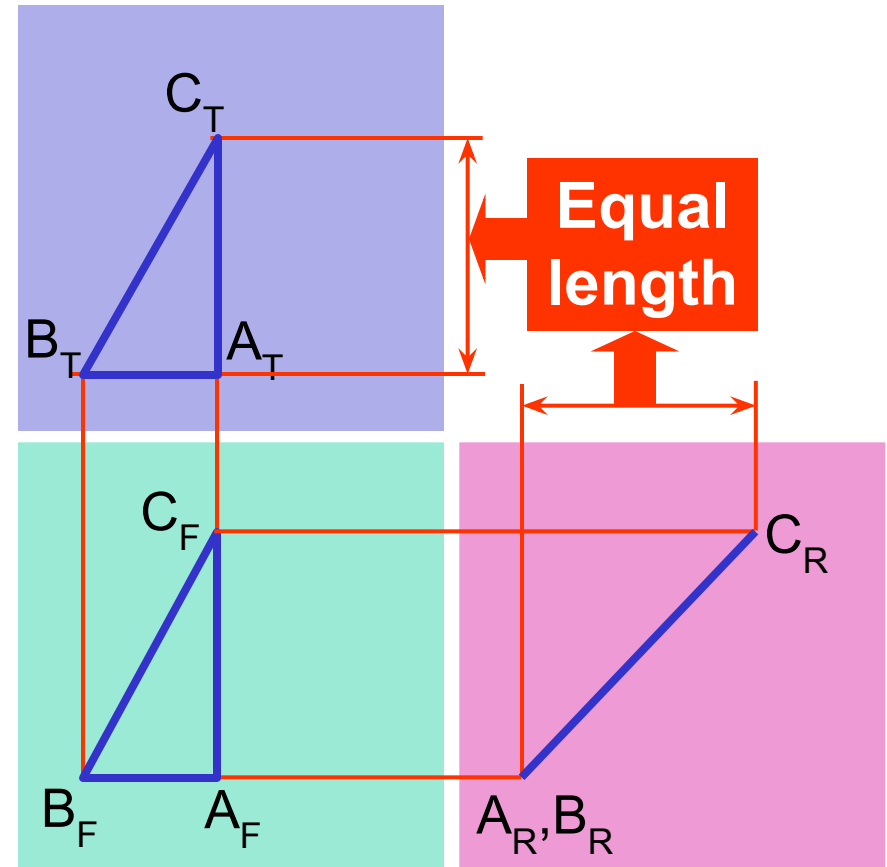
PROJECTION OF PLANE

Foreshortened



Foreshortened

Edge

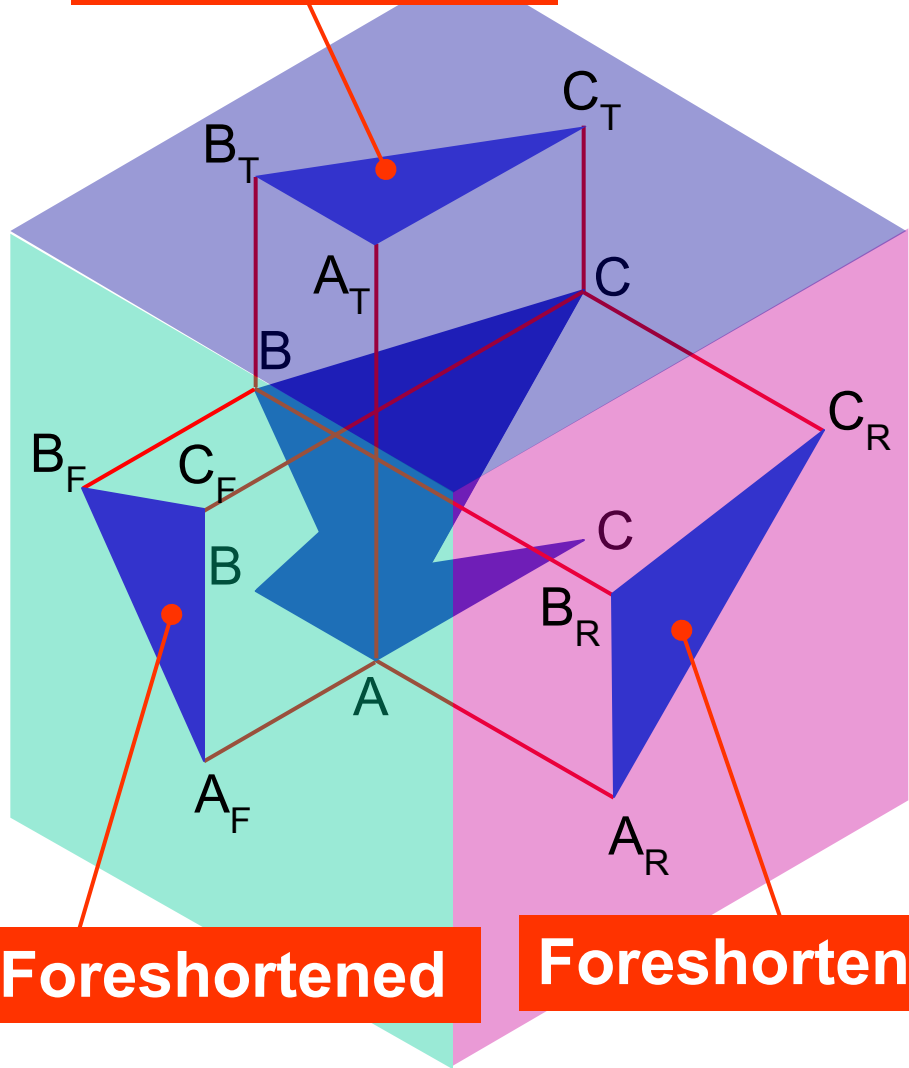


Equal length

INCLINED PLANE

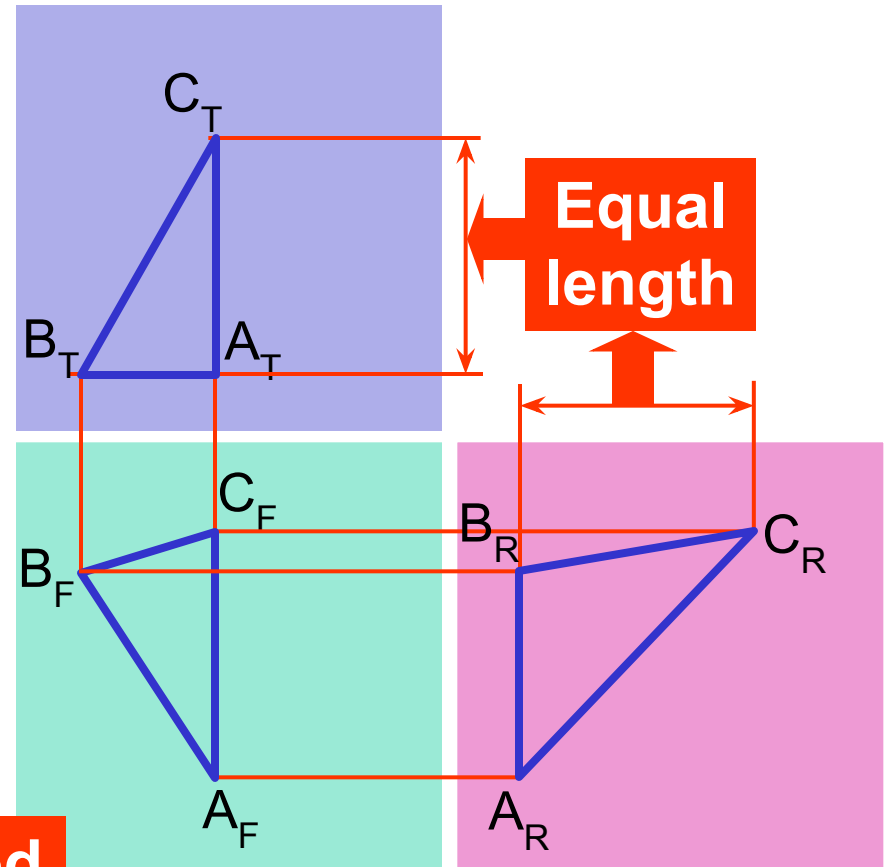
PROJECTION OF PLANE

Foreshortened



Foreshortened

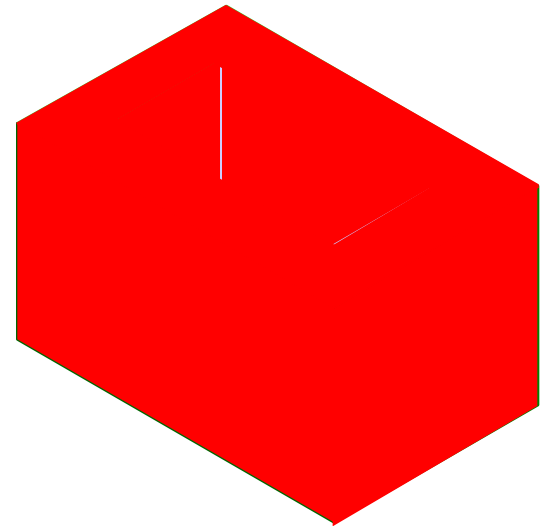
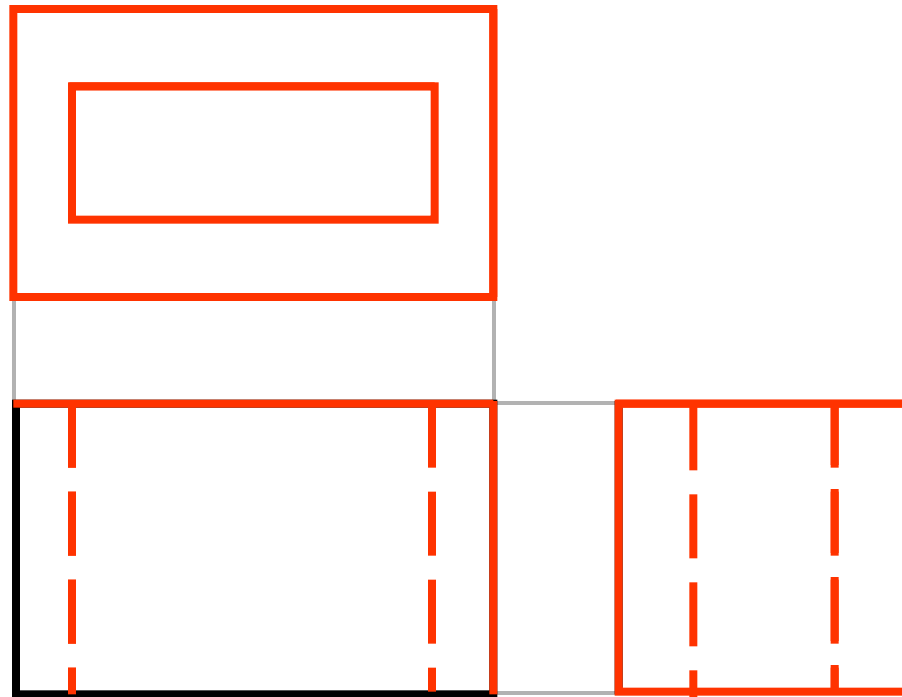
Foreshortened



OBLIQUE PLANE

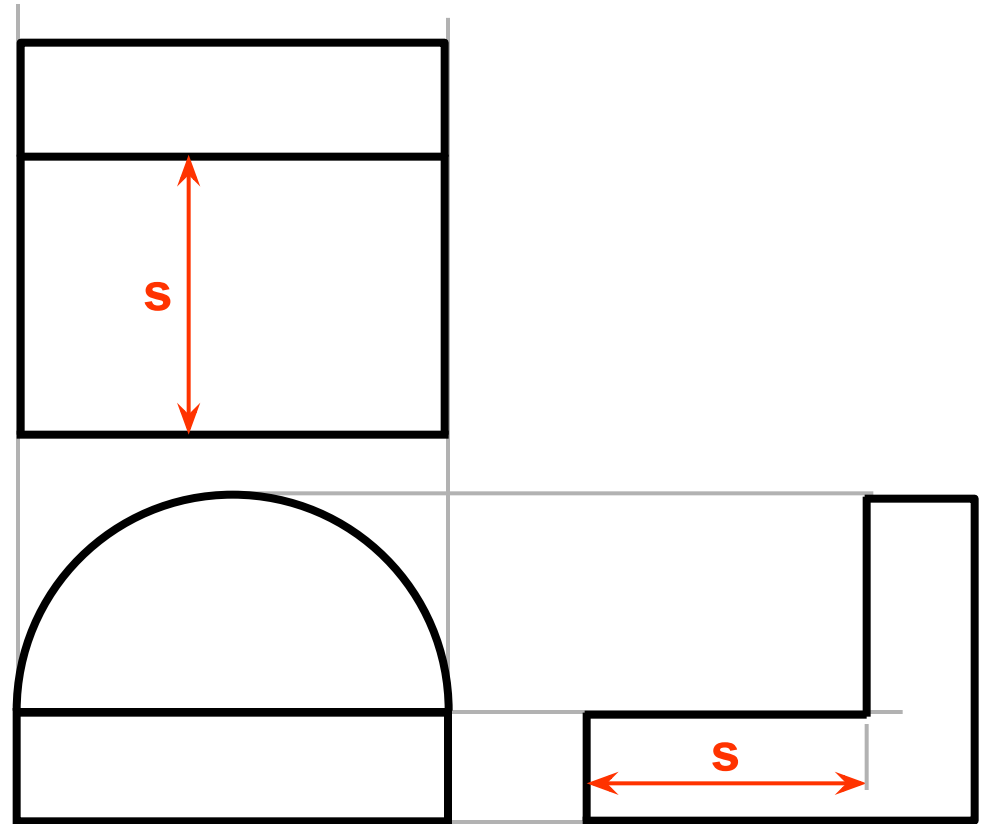
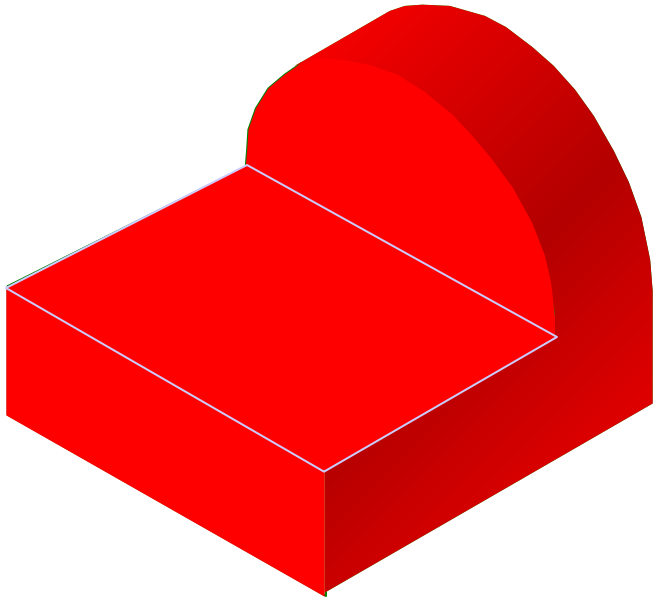
PROJECTION OF OBJECT

The views are obtained by projecting all object features to the picture plane.

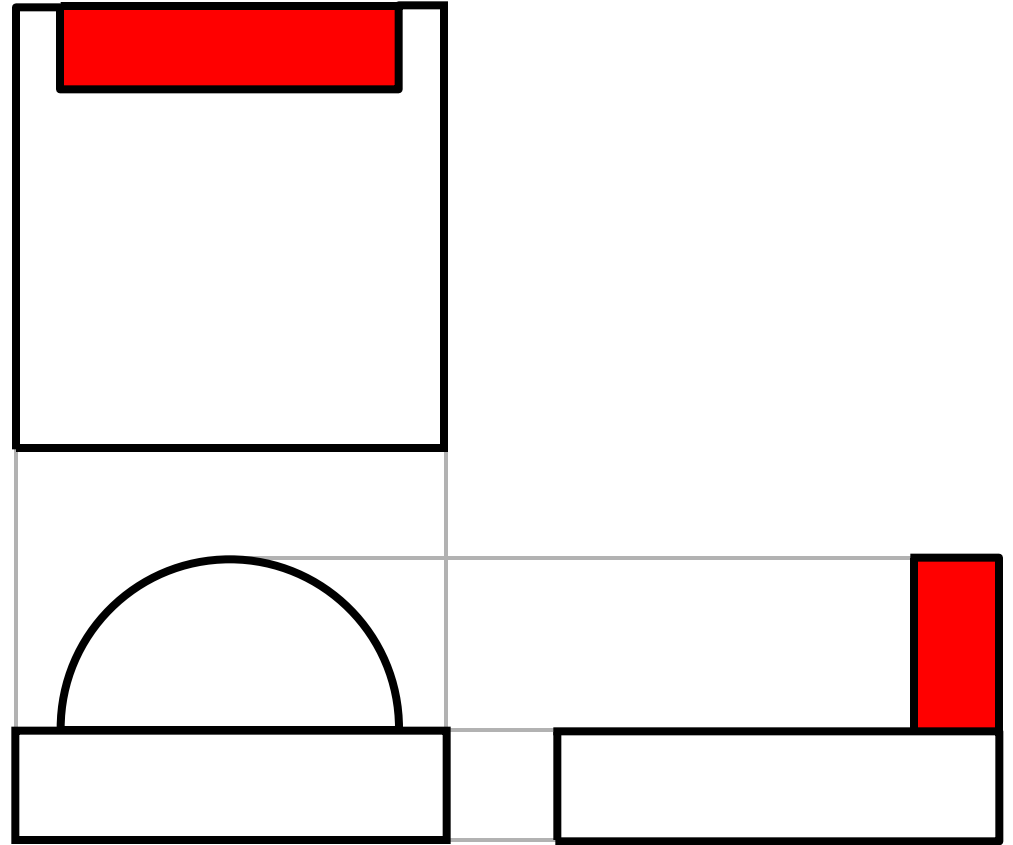
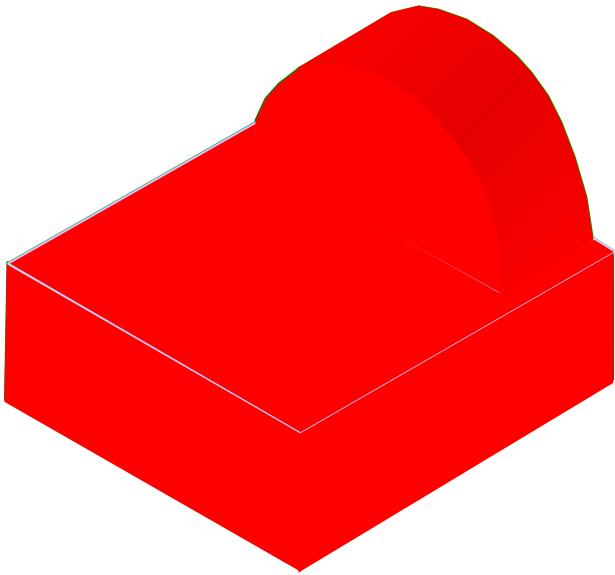


You have to project the remaining surfaces which are invisible too !

PROJECTION OF OBJECT



PROJECTION OF OBJECT





Line Convention



LINE CONVENTION

- Precedence of coincide lines.
- Hidden line drawing.
- Center line drawing.

PRECEDENCE OF LINE

Order of
importance

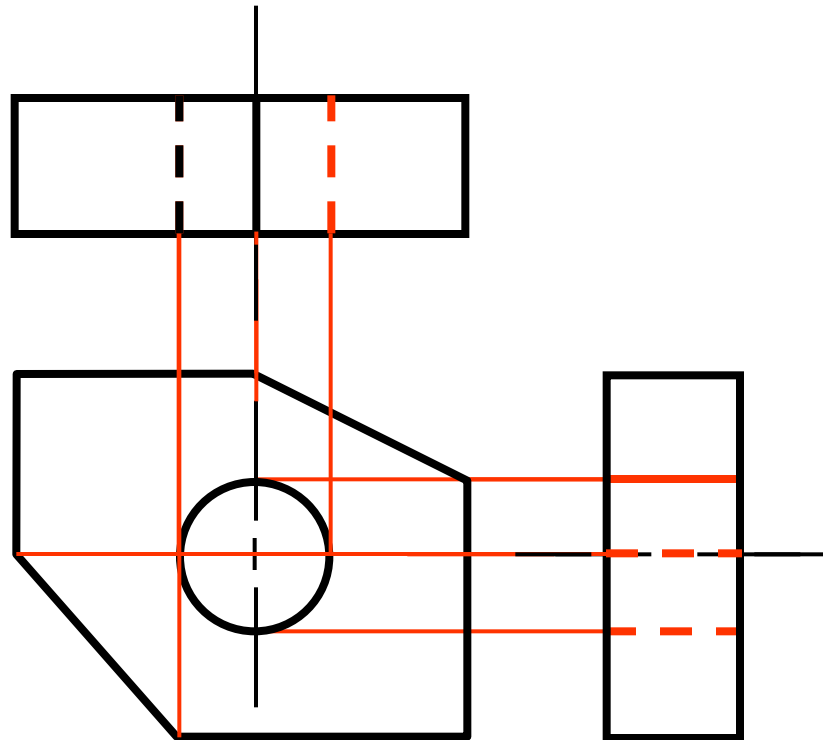
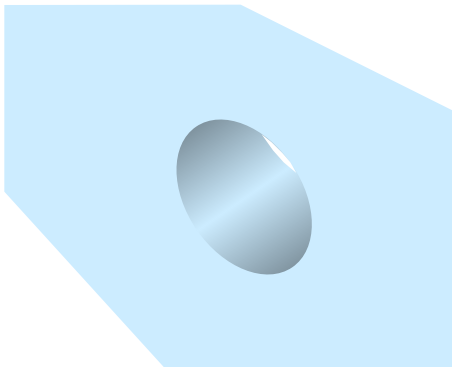
Visible
line



Hidden
line

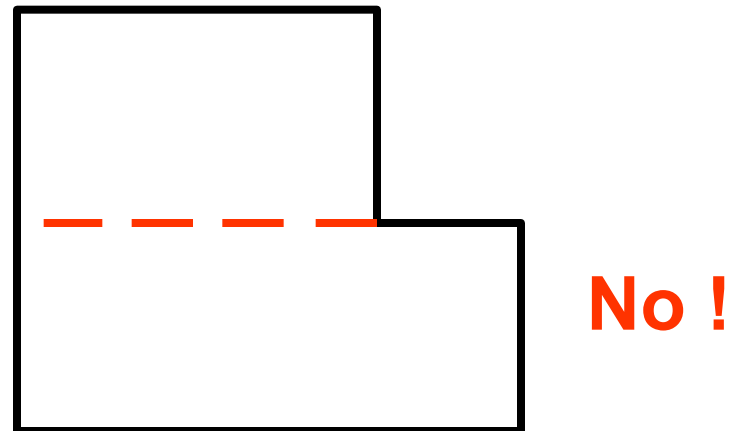
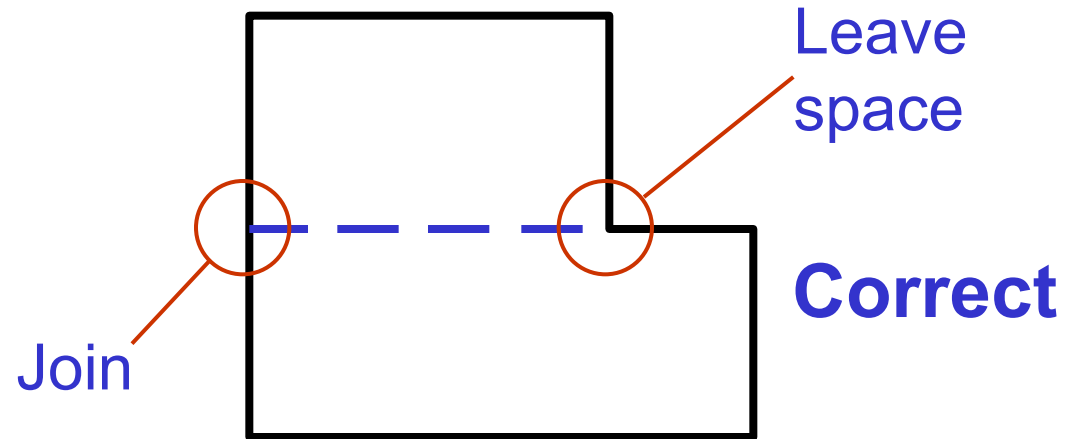
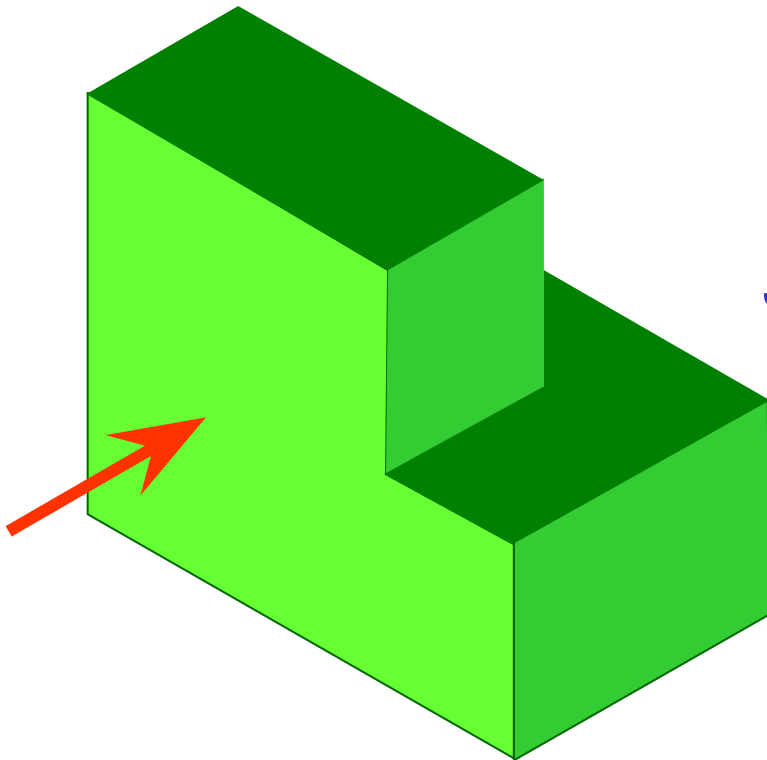


Center
line



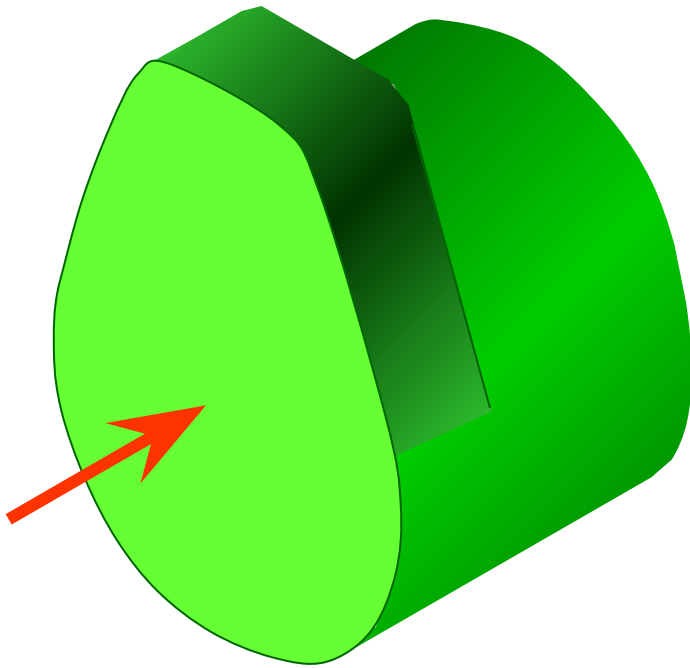
HIDDEN LINE PRACTICE

- Hidden line should join a visible line, **except** if extended from the visible line.



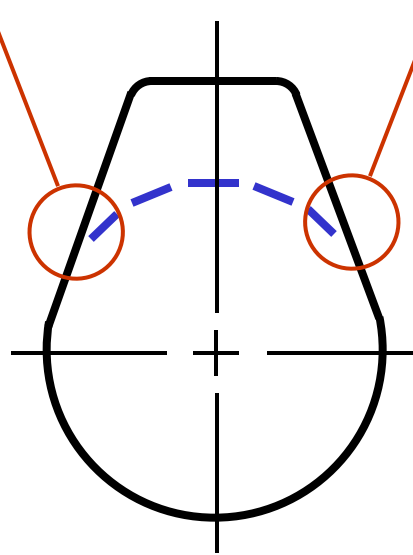
HIDDEN LINE PRACTICE

- Hidden line should join a visible line, **except** if extended from the visible line.

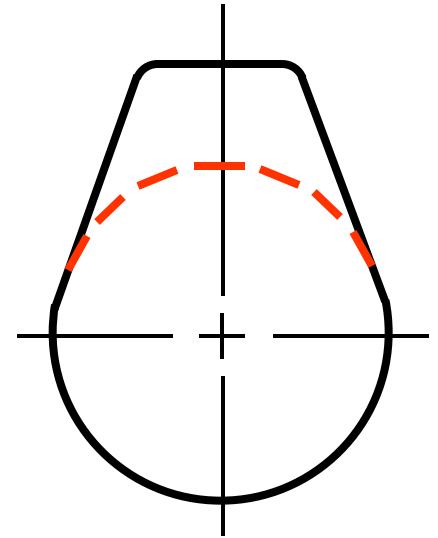


Leave space

Leave space



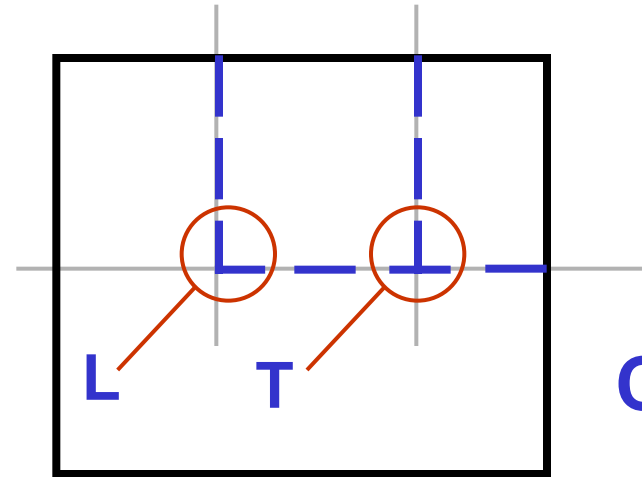
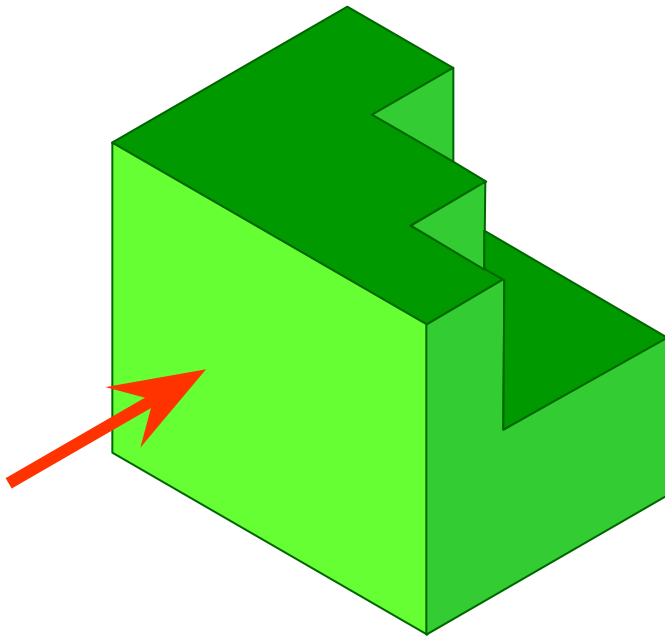
Correct



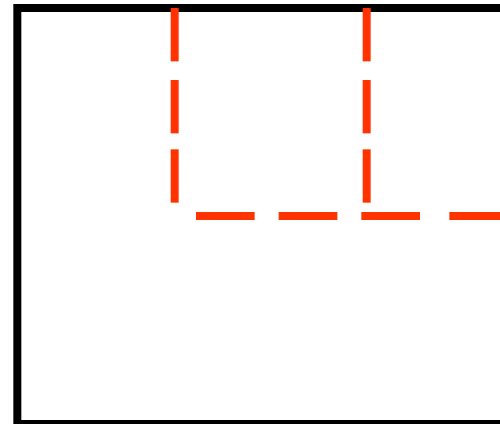
No !

HIDDEN LINE PRACTICE

- Hidden line should intersect to form **L** and **T** corners.



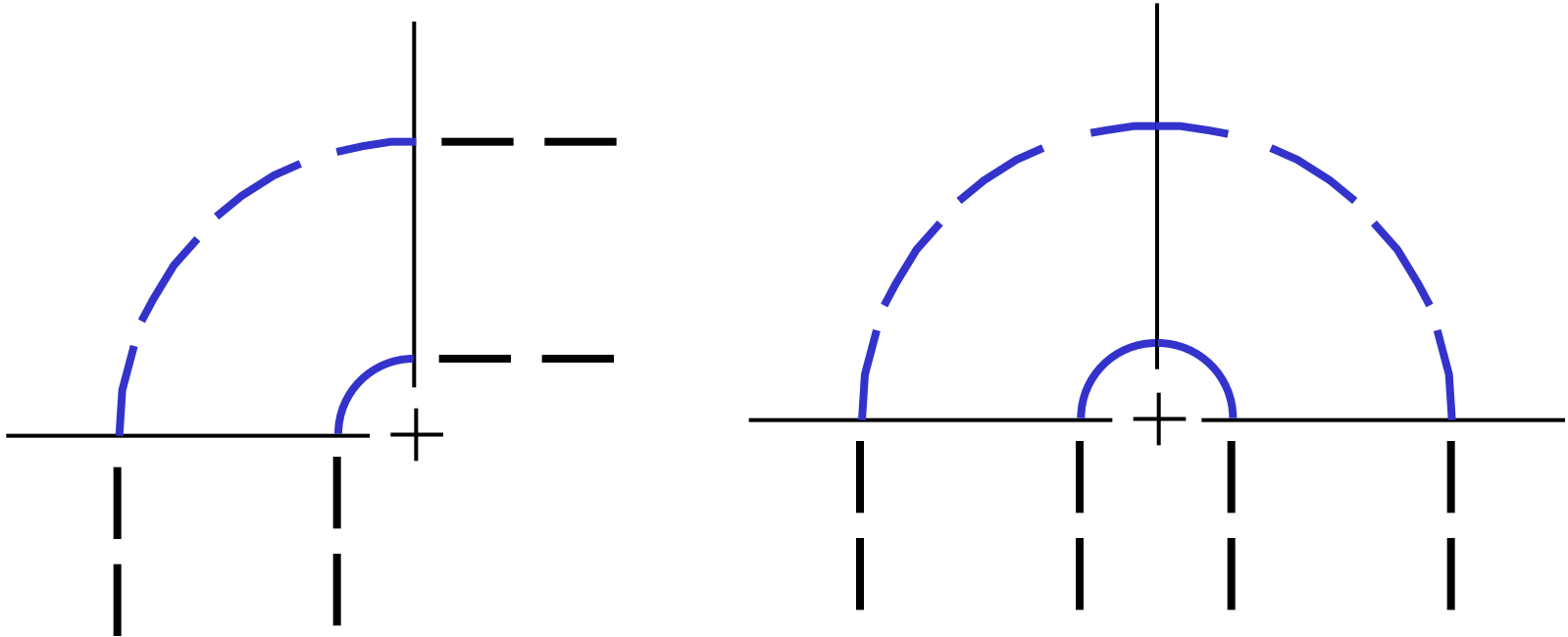
Correct



No !

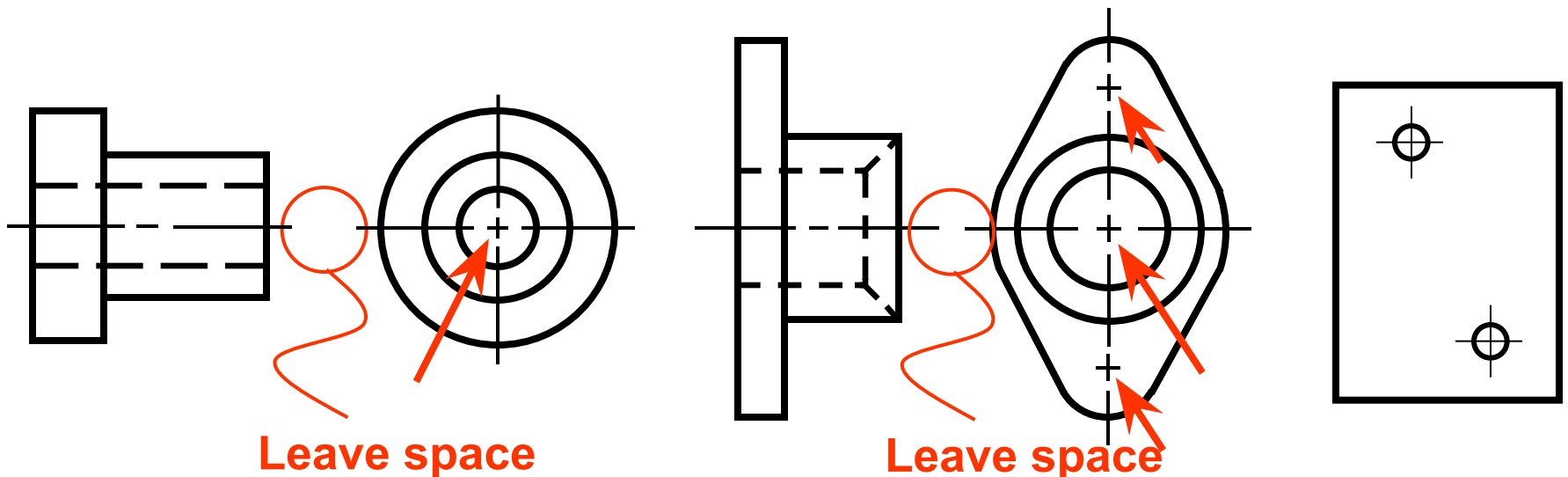
HIDDEN LINE PRACTICE

- Hidden arcs should start on a center line.



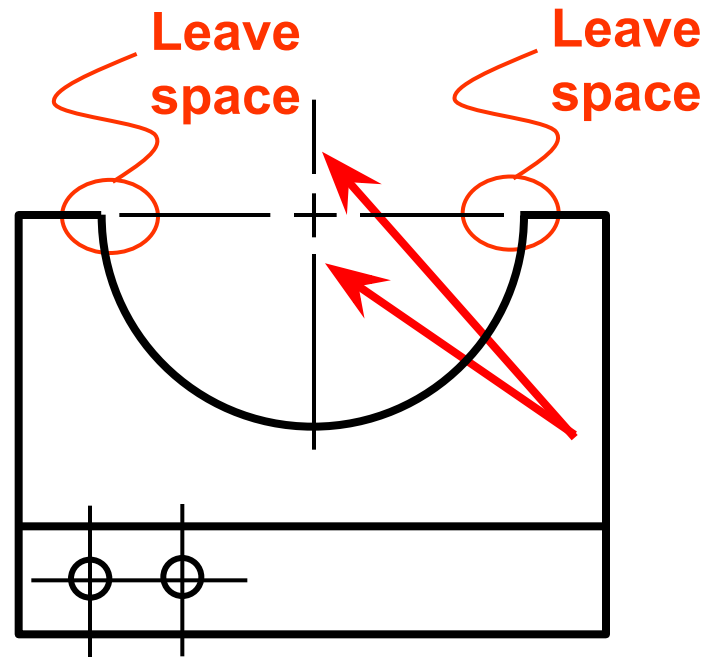
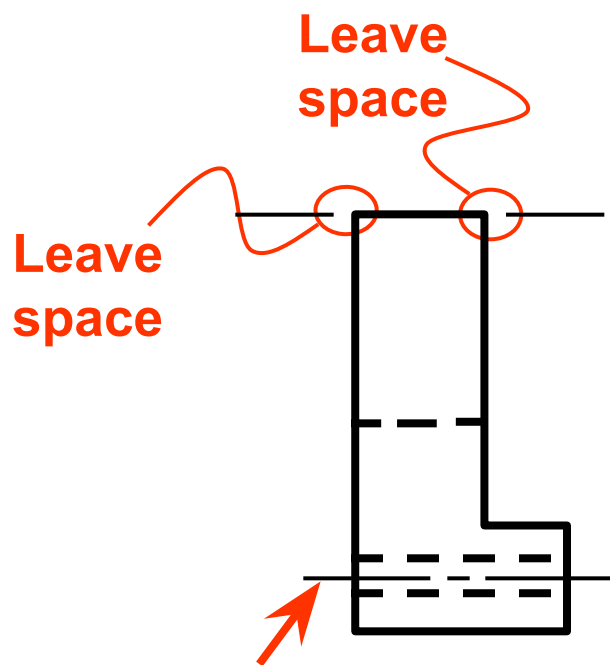
CENTER LINE PRACTICE

- In circular view, short dash should cross at the intersections of center line.
- For small hole, center line is presented as thin continuous line.
- Center line should not extend between views.



CENTER LINE PRACTICE

- Leave the gap when centerline forms a continuation with a visible or hidden line
- Center line should always start and end with long dash.



Discussion